Dynamics: local stability, phase portraits and bifurcations

September 10, 2026

Abstract

A simulations answers one exact question about **one** model, **one** parameter point, and **one** initial condition. The class builds the map **structure** $\rightarrow$ **function**

Section 1

Why we analyze if we can already simulate?

A model that we already master

Concepts

- $\dot{x} = \Gamma v(x)$ is the general form of a reaction network (lecture 3): $\Gamma =$ stoichiometric matrix, $v(x) =$ flux vector.
- **Constitutive** protein: produced at a constant rate μ (no regulation) and removed at first-order rate λx (degradation + dilution because the cell grows and "partitions" the protein among daughters.

$$\dot{x} = \mu - \lambda x \tag{1}$$

- **Fixed point (steady state).** Where the net rate is zero

$$x^* = \frac{\mu}{\lambda} \tag{2}$$

- Solution:

$$x(t) = x^* + (x_0 - x^*)e^{-\lambda t} \tag{3}$$

- The **relaxation time** is $\tau = 1/\lambda$.

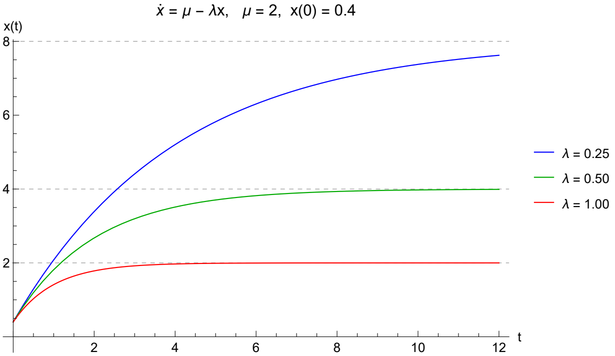

Figure: The three curves $x(t)$ with $\lambda = 0.25, 0.5, 1.0$; the dashed horizontal lines are $\mu/\lambda = 8, 4, 2$.

If λ goes from 0.5 to 1.0, where does the steady level move, and by what factor does the relaxation time change?

The two limits

1. The model was built by *compressing*. What was compressed **is not in the equations**, so the simulation cannot report it.
2. 10 parameters $\times 10$ values $= 10^{10}$ runs, and we still haven't sampled initial conditions. We need statements that cover infinite families at once.

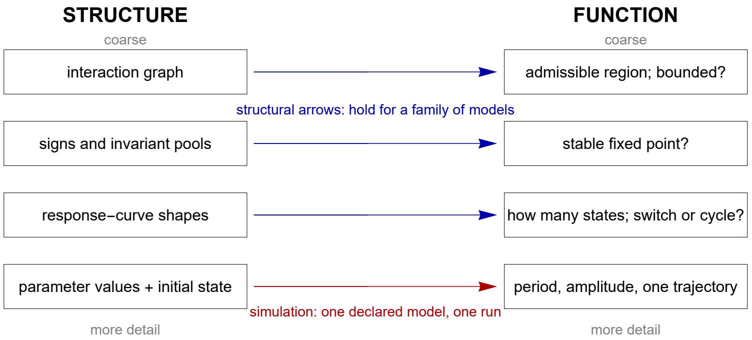

Figure: a structural statement declares what information it uses and over what family of models it is valid.

Section 2

One dimension, one fixed point is one crossing

Addition minus removal

For a concentration $x \geq 0$ one can always write

$$\dot{x} = \underbrace{f^+(x)}_{\geq 0} - \underbrace{f^-(x)}_{\geq 0} = f(x) \tag{4}$$

For example, $f^+ = \mu$, $f^- = \lambda x$

The slope condition

At a fixed point $f^+(x^*) = f^-(x^*)$, **Question: Is it stable?**

1. First-order Taylor around x^*

$$f(x) \approx f'(x)(x - x^*) \tag{5}$$

2. Suppose $f'(x^*) < 0$. Then
   - If $x > x^*$ (you are to the right): $x - x^* > 0$, so $\dot{x} \approx$ (negative)(positive) $< 0 \rightarrow$ **you move left, back**
   - If $x < x^* : \dot{x} > 0 \rightarrow$ **you move right, back**

Both directions return: **stable** in terms of the curves

The three fields, solved

- **Decay:** $\dot{x} = -x$; $x = x_0 e^{-t}$: stable
- **Non-hyperbolic growth:** $\dot{x} = x^2$; $x = x_0/(1 - x_0 t)$: stable $\rightarrow$ left, unstable $\rightarrow$ right
- **Bistable switch:** $\dot{x} = -(x-1)(x-2)(x-3)$; classification with derivative
  1. $f'(1) = -2 < 0$: stable
  2. $f'(2) = 1 > 0$: unstable
  3. $f'(3) = -2 < 0$: stable

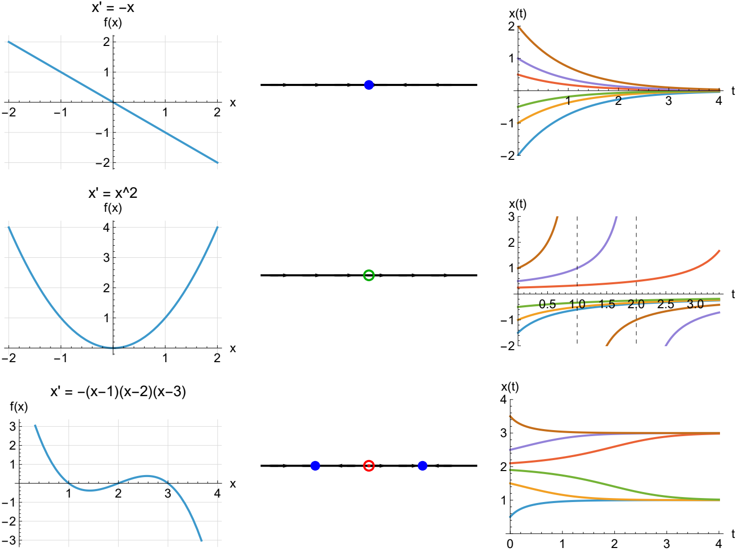

Figure: f(x), the phase line and the traces $x(t)$.

The first impossibility theorem

Between two consecutive roots, f does not change sign (it's) continuous $\rightarrow$ the trajectory is **monotone** there. To return, it would have to reverse direction, which requires passing through a zero of f (*a fixed point*); but there it stops forever (uniqueness of solutions).

> ***A scalar autonomous system can switch, but it can not oscillate sustainably.***

Section 3

To remember, cross three times

Memory as a design problem

“Memory” $=$ two persistent stable states. By the previous theorem, between two stables states there must be and unstable one: the sequence is crossing: **stable–unstable–stable**.

The self regulated gene, piece by piece

$$\frac{dx}{d\tau} = \underbrace{0.15}_{\text{basal}} + \underbrace{\frac{3x^4}{1+x^4}}_{\text{Hill activator}} - \underbrace{x}_{\text{dilution}} \tag{6}$$

- **0.15:** basal (the gene "leaks" even when off) – keeps the low state away from zero.
- **Hill term:** the protein activates its own production (positive feedback).
- **-x:** first order removal.

The three crossings – calculation step by step

Solving $0 = 0.15 + \frac{3x^4}{1+x^4} - x$ we find the roots $\approx 0.152, 0.681, 3.119$

Stability with slope condition:

$$(f^+)'(x) = \frac{d}{dx}\left(0.15 + \frac{3x^4}{1+x^4}\right) = \frac{12x^3}{(1+x^4)^2} \tag{7}$$

Since $(f^-)'(x) = 1$, the point is stable if $(f^+)'(x) < 1$:

$x^* = 0.152 : (f^+)' \approx 0.042 \to f' \approx -0.96 < 0 \to$ **stable** low state
$x^* = 0.681 : (f^+)' \approx 2.57 \to f' \approx 1.57 > 0 \to$ **unstable** threshold
$x^* = 3.119 : (f^+)' \approx 0.040 \to f' \approx -0.96 < 0 \to$ **stable** high state

Interpretation: a transient signal is "remembered" only if it pushes x above the threshold 0.681 into the basin of the high state.

The logic of shape and the “desing check”

Three necessary ingredients for memory (3 crossings):

- the positive feedback (goes up)
- sufficient slope (locally steeper than removal)
- saturation (flattens up and removal wins again)

1. **If slope $\leq$ 1 everywhere:** only 1 crossing $\rightarrow$ no memory.
2. **If no saturation:** curve never comes back down $\rightarrow$ no high stable state (runaway).

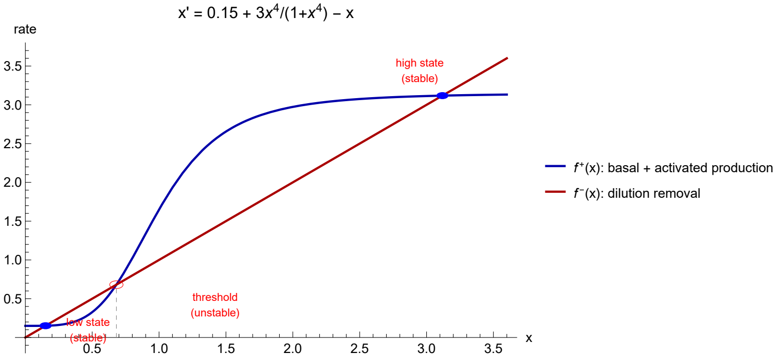

Figure: The two lines with the three crossings marked (blue=stable, red=threshold)

Section 4

When two curves touch, a switch is born

Concept of bifurcation

A **bifurcation** is a qualitative change in the organization of *all* trajectories as a parameter varies (number/stability of fixed points) – not just “a curve that bends”.

The saddle–node formal form

Let r be a control parameter and x a local displacement from the collision point. The **normal form** of the saddle-node bifurcation is

$$\dot{x} = r - x^2 \tag{8}$$

Fixed points: $\pm\sqrt{r}$ exist only if $r \geq 0$.

- $r < 0$: no roots.
- $r = 0$: $x^2 = 0$ at $x = 0$ (double root)
- $r > 0$: 2 roots. Stability at $\frac{df}{dx} = -2x$

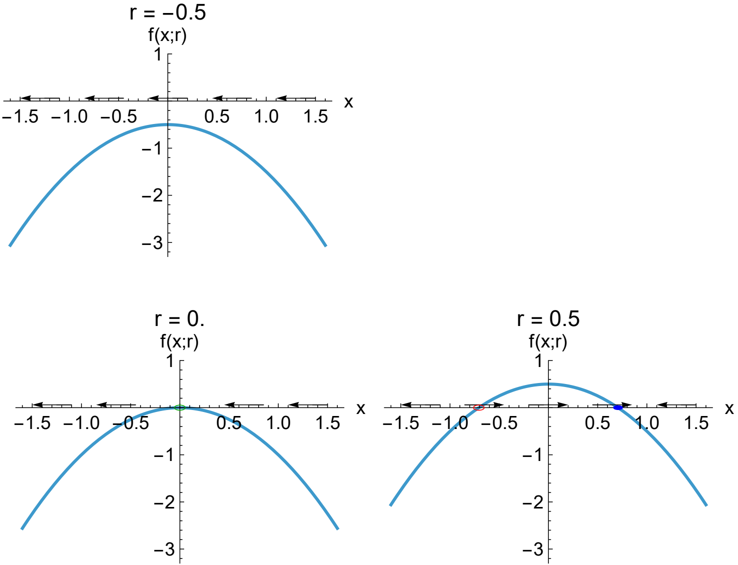

Figure: As r varies, a stable–unstable pair is **born** (or dies) at $r = 0$: hence "saddle–node" and "fold".

In the addition–removal language, the birth of the pair occurs when the curves **touch**: they equal values and slopes

$$f^{+} = f^{-} \text{ (crossing) and } (f^{+})' = (f^{-})' \text{ (tangency i.e. } f' = 0)$$

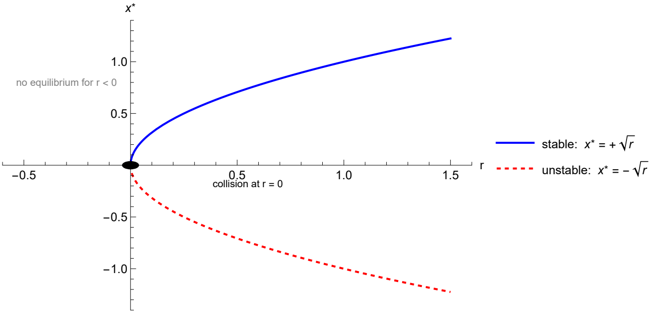

Figure: Bifurcation diagram.

Hysteresis and critical slowing down

Hysteresis: a sigmoidal switch has two folds; increasing the input destroys one state at one fold and decreasing destroys the other at the other fold $\rightarrow$ the threshold depends on the direction and history.
Near the stable root, let $u = x - \sqrt{r}$ be a small displacement

$$\dot{u} \approx -2\sqrt{r}u \Longrightarrow u(t) = u(0)e^{-2\sqrt{r}t} \tag{9}$$

The decay is exponential with rate $2\sqrt{r}$

$$\tau_{\text{relax}} = \frac{1}{2\sqrt{r}t}$$

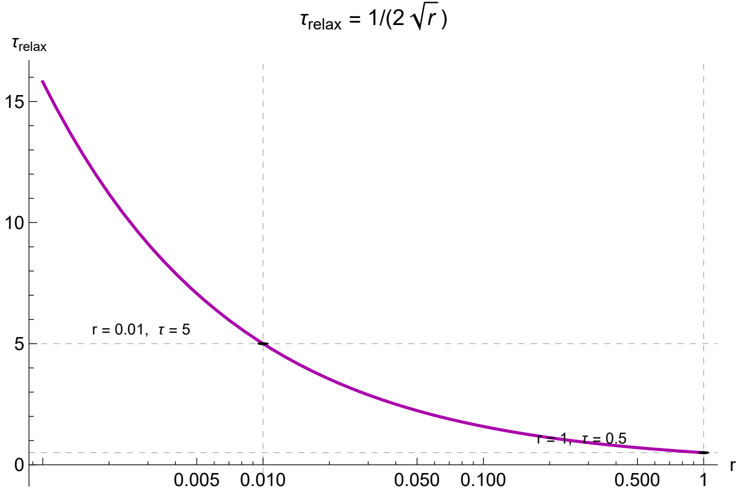

Figure: τ_{relax} vs r on logarithmic scale.

Section 5

Two dimensions, the toggle

Nullclines

Graphical methods are based on the construction of what is called a *direction field* for $f' = f(x, y)$. The **isoclines** are the line segments all have the same slope m. The figure below shows a direction field for the equation $y' = y - x$. The $m = 0$ isocline is called a **nullcline** (direction field is horizontal).

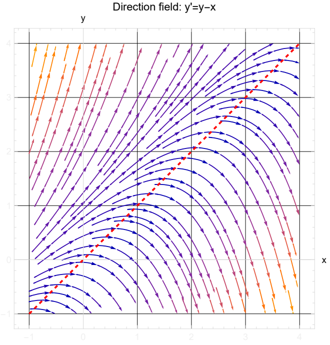

Figure: The direction of the field is horizontal along the nullcline.

The network rotation

- $G_X \rightsquigarrow G_X + X$: the gene G_X produces protein X without being consumed.
- $X \to \emptyset$: the protein degradates/dilutes
- $Y \dashv G_X$: protein Y **represses** gene G_X ($\dashv$ is inhibition)

Rapid equilibrium model: the promoter is either free (G_X) or occupied by the repressor $C_{G_X Y}$ with occupancy ratio

$$\frac{C_{G_X Y}}{G_X} = \left(\frac{Y}{K_Y}\right)^{n_Y}$$

and conservation of a total pool: $q_{G_X} = G_X + C_{G_X Y}$.

Solving for the free fraction:

$$\frac{G_X}{q_{G_X}} = \frac{G_X}{G_X + C_{G_X Y}} = \frac{1}{1 + C_{G_X Y}/G_X} = \frac{1}{1 + (Y/K_Y)^{n_Y}} \quad (10)$$

The more repressor Y, the smaller the free fraction $\rightarrow$ the less X production.
If the lumped production is proportional to the free promoter, with maximum production α_X and removal $\delta_X X$:

$$\dot{X} = \frac{\alpha_X}{1 + (Y/K_Y)^{n_Y}} - \delta_X X, \quad \dot{Y} = \frac{\alpha_Y}{1 + (X/K_X)^{n_X}} - \delta_Y Y \quad (11)$$

The symmetric case and its fixed points

Scaling concentrations are:

$$\dot{x} = \frac{4}{1+y^2} - x, \quad \dot{y} = \frac{4}{1+x^2} - y \tag{12}$$

Nullclines: $x = \frac{4}{1+y^2}$ and $y = \frac{4}{1+x^2}$
Fixed points: $(1.38, 1.38), (0.27, 3.73), (3.73, 0.27)$

The Jacobian and eigenvalues

$$J(x,y) = \begin{bmatrix} \frac{\delta F_1}{\delta x} & \frac{\delta F_1}{\delta y} \\ \frac{\delta F_2}{\delta x} & \frac{\delta F_2}{\delta y} \end{bmatrix} = \begin{bmatrix} -1 & \frac{-8}{(1+y^2)^2} \\ \frac{-8}{(1+x^2)^2} & -1 \end{bmatrix} \tag{13}$$

Evaluated at symmetric point 1.3788

$$J = \begin{bmatrix} -1 & -1.3106 \\ -1.3106 & -1 \end{bmatrix} \tag{14}$$

Eigenvalues: $\det(J - \lambda I) = 0$:
$\lambda_1 = 0.3106 > 0$, $\lambda_2 = -2.3106 < 0 \rightarrow$ **saddle**
Evaluated at (0.2679,3.7321)

$$J = \begin{bmatrix} -1 & -0.0359 \\ -6.9644 & -1 \end{bmatrix} \tag{15}$$

Eigenvalues: $\lambda_1 = -1.5 > 0$, $\lambda_2 = -0.5 < 0 \rightarrow$ **stable**

The separatrix

By symmetry ($x \longleftrightarrow y$), the diagonal $y = x$ is invariant. The stable manifold of the saddle is that diagonal: it separates the basins of "x high / y low" and "x low / y high"

fixed point	eigenvalues	type	interpretation
$(0.27,\ 3.73)$	$-0.5,\ -1.5$	stable node	low X / high Y
$(1.38,\ 1.38)$	$+0.31,\ -2.31$	saddle	threshold state
$(3.73,\ 0.27)$	$-0.5,\ -1.5$	stable node	high X / low Y

Table: Fixed points and eigenvalues.

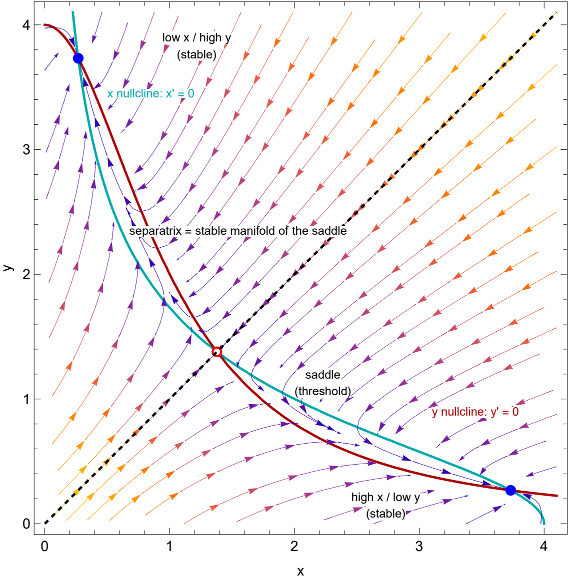

Figure: Nullclines (turquoise/red), flow field, the three fixed points, and the dashed separatrix.

Section 6

The frontier of dimension

The packing argument. Trajectories cannot cross (uniqueness), but the power of that rule depends on dimension:

- **1D:** ordered, monotone between fixed points.
- **2D:** an isolated closed orbit separates the plane into inside/outside. **Poincaré-Bendixon:** if a bounded, invariant region of the plane contains no fixed points, it contains a periodic orbit. The catalog is short: nodes, saddles, spirals, centers, isolated cycles
- **3D:** a curve pass "over" itself; loops, braids and chaos fit.

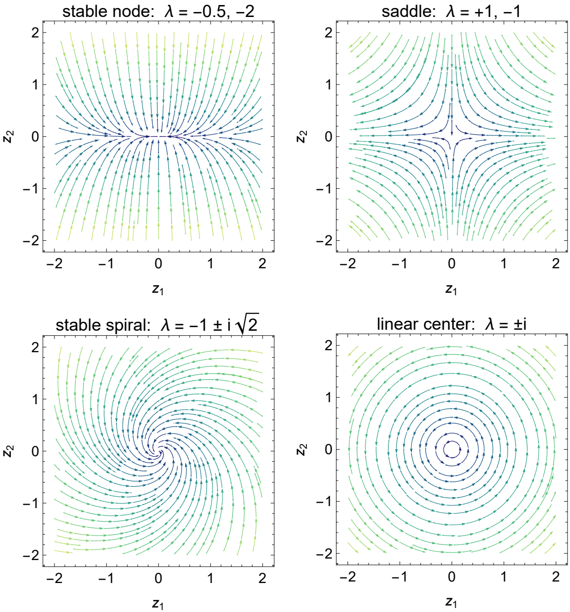

Figure: The four modal portraits.

Why chaos needs 3 dimensions?

- 1D (a line). Trajectories can only go to the left or right.
- 2D (a plane). A trajectory can rotate, but **cannot cross itself**.
- 3D (in space). Now a trajectory can **pass above or below** itself, like a knot or a braid. *It can twist without touching itself, and that allows chaos.*

The Lorenz attractor (the classic example)

- It's 3 equations that model the atmosphere (air convection).
- Its trajectories never repeat, but they never escape a butterfly-shaped region.
- The curve swings around one "wing," then passes over to the other, and so on forever

The Lorenz attractor

$$\frac{dx}{dt} = \sigma(y - x) \tag{16}$$

$$\frac{dy}{dt} = x(\rho - z) - y \tag{17}$$

$$\frac{dz}{dt} = xy - \beta z \tag{18}$$

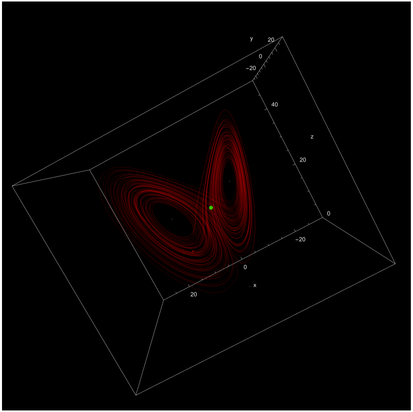

Figure: Classical parameters: $\sigma = 10$, $\rho = 28$, $\beta = 8/3$.

Section 7

Near a fixed point every network is linear

Linearization

Taylor Series Expansion

$$F(x) \approx \underbrace{f(x_0) + \left.\frac{df}{dx}\right|_{x_0} (x - x_0)}_{\text{linear part}} + \left.\frac{d^2 f}{dx^2}\right|_{x_0} (x - x_0) + \dots \tag{19}$$

The linear system $\dot{x} = Ax$ assumes a solution in the form $x(t) = e^{At}x(0)$.

Eigenvalues and Stability

Let $\lambda = a + ib$

$$e^{\lambda t} = e^{(a+ib)t} = e^{at}(\cos(bt) + i\sin(bt))$$

If $a > 0$ solution blow up.
If $a < 0$ solution stabilizes.
The system is stable is **all** eigenvalues are **negatives**.
A is Hurtwitz if every eigenvalue has a negative real part.

Hurtwitz criterion

For 2×2, the characteristic polynomial is

$$\lambda^2 - \tau\lambda + \Delta = 0, \qquad \lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

with $\tau = \operatorname{tr} A = \lambda_1 + \lambda_2$ and $\Delta = \det A = \lambda_1 \lambda_2$.
Proof of the criterion $\tau < 0$ and $\Delta > 0$ by cases:

- $\Delta < 0$: real eigenvalues of opposite sign $\to$ one positive $\to$ **saddle, always unstable**.
- $\Delta > 0$, $\tau^2 \geq 4\Delta$ (real): same sign as their sum $\tau \to$ both negative $\iff \tau < 0$ (**stable node**; if $\tau > 0$, unstable node).
- $\Delta > 0$, $\tau^2 < 4\Delta$ (complex conjugates): real part $= \tau/2 \to$ decay $\iff \tau < 0$ (**stable spiral**; if $\tau > 0$, unstable spiral).
- $\tau = 0$, $\Delta > 0$: $\lambda = \pm i\sqrt{\Delta}$ pure imaginary $\to$ **linear center**.

The mass–spring–damper system:

$$m\ddot{x} + b\dot{x} + kx = 0 \tag{20}$$

where m is the mass, b is the friction constant, k is the spring coefficient.

$$\frac{d}{dt}\begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix}\begin{bmatrix} x \\ \dot{x} \end{bmatrix} \tag{21}$$

Roots are:

$$x^* = \frac{-b \pm \sqrt{b^2 - 4mk}}{2a} \tag{22}$$

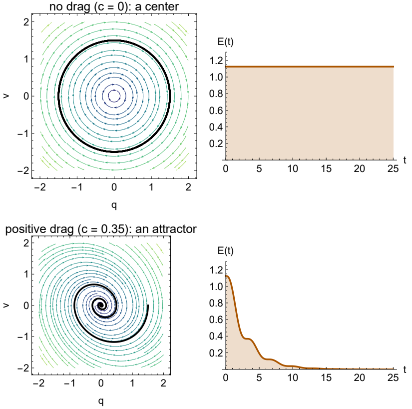

Figure: Up: without drag, closed orbits and constant E. Down: with drag, energy falls and every trajectory converges.

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